Automated Glacial Lake Mapping and Hazard Assessment in the Hindu Kush–Himalaya Using Deep Learning and PlanetScope Imagery

Muhammad Bilal
Information Technology University
Lahore, Pakistan
msds24036@itu.edu.pk

Muhammad Adnan Siddique
Information Technology University
Lahore, Pakistan
adnan.siddique@itu.edu.pk

Ehtasham Naseer
Information Technology University
Lahore, Pakistan
phdee21002@itu.edu.pk

***Abstract*—We present an end-to-end deep learning framework for automated glacial lake mapping and hazard assessment across the Hindu Kush–Himalaya using PlanetScope imagery. CNN- and transformer-based segmentation models were trained on RGB and multi-spectral inputs to generate a consistent 2025 glacial lake inventory. RGB-only models achieved the best performance, with MANet and U-Net++ attaining IoU values above 0.67. Comparative analysis between 2020 and 2025 inventories reveals increasing lake counts but a net reduction in total lake area. A multi-factor hazard index identifies nearly one-quarter of mapped lakes as potentially dangerous, enabling downstream population exposure assessment.**

***Index Terms*—Glacial lakes, deep learning, semantic segmentation, PlanetScope imagery, hazard assessment, potentially dangerous glacial lakes (PDGLs), Hindu Kush–Himalaya.**

I. Introduction

Glacial lakes in high mountain regions are rapidly evolving under climate change, increasing the risk of glacial lake outburst floods (GLOFs). Accurate, up-to-date lake inventories are essential for hazard assessment but are limited by manual mapping and inconsistent temporal coverage. Recent advances in deep learning offer scalable solutions for automated lake detection from high-resolution satellite imagery. However, the impact of model architecture and spectral configuration on glacial lake segmentation remains insufficiently explored. This study addresses these gaps by developing an operational deep learning pipeline for lake mapping, change analysis, and PDGL identification across the Hindu Kush–Himalaya.

II. Methodology

A. Overview

We created an end-to-end framework for automated glacial lake segmentation training and inference from PlanetScope 4-band visual imagery and hazard assessment using multi-factor PDGL identification framework.

B. PlanetScope Imagery Retrieval, Pre-Processing and Dataset Generation

For training we used HKH-2020 glacial lake inventory [1]. The corresponding scenes for the lakes were downloaded as 4-band visual and 8-band SR products with cloud cover $< 20\%$. Each scene was tiled into 512×512 px geo-referenced patches with 20% overlap to keep spatial context. Tiles with less than 60% valid pixels were removed to eliminate mostly empty tiles that usually comes from irregular shaped scene edges.

HKH-2020 lake polygons intersecting each tile were rasterized into binary masks ($1\ =$lake, $0\ =$non-lake). After removing tiles with no lake pixels, a refined dataset of 750 pairs was obtained for 4-band visual product and 850 for 8-band SR product.

C. Model, Training Setup and Inference

Spectral Configurations: We evaluated multiple input-band configurations. (1) RGB only: compatible with ImageNet pre-trained encoders. (2) Full 8-band SR: modifying first-layer convolutions to accept 8 channels. (3) RGB + NIR + NDWI: NDWI computed as

$$\mathrm{NDWI} = \frac{G - NIR}{G + NIR},$$

appended as a fifth channel.

Segmentation Models: We trained both CNN- and transformer-based models.

1) CNN Models: U-Net [2], U-Net++ [3], U-Net++ with Attention Decoder [4], DeepLabv3+ [5], and MANet [6] were implemented using Segmentation Models PyTorch (SMP). Backbone encoders included ResNet34, ResNet50, timm-EfficientNet-B0 (EfficientNet-B0 for U-Net), and MobileNet-V2.

2) Transformer Models: Three transformer-based architectures were evaluated. TransUNet [7] with Optuna hyperparameter search (attention heads $\in \{4, 8\}$, depth $\in \{2, 4, 6\}$, MLP ratio $\in \{2, \ldots, 5\}$), Swin U-Net [8] with MiT encoders (mit-b0, mit-b1, mit-b2, mit-b3) and SegFormer [9] with encoders ResNet34, ResNet50, timm-EfficientNet-B0, and MobileNet-V2.

Training Setup: All models were trained using a unified framework. BCE + Dice was used as loss function. Adam optimizer was used with, learning rate 1×10^{-3} (for U-Net, U-Net++, U-Net++ with Attention Decoder, DeepLabv3+, Swin U-Net, SegFormer) and 1×10^{-4} (for MANet and

TransUNet). We kept batch size 2 and trained for 10 epochs for screening; 20 for top-performing RGB models (U-Net++, MANet, SegFormer). Metrics used were IoU, Precision, Recall and F1-score.

Inference Pipeline: An operational pipeline was developed to generate GIS-ready lake polygons. Using the pipeline, we derived a lake inventory for 2025.

D. Lake Stats Computation

Lake Area Computation: Total lake area for each inventory year was obtained by summing the areas of all individual lake polygons. Lake count (N) was defined as the total number of polygons in each inventory.

Lake Size Classification: To analyze size-dependent patterns, lakes were grouped into predefined size classes based on surface area: < 0.01 km^2, 0.01–0.05 km^2, 0.05–0.1 km^2, 0.1–0.5 km^2, 0.5–1 km^2, 1–5 km^2, 5–10 km^2, and > 10 km^2. For each size class and inventory year, the number of lakes and the cumulative lake area were computed.

Elevation Attribution: Lake elevation was estimated using the Shuttle Radar Topography Mission (SRTM) 1 arc-second ($\sim$30 m) digital elevation model (DEM). Multiple SRTM tiles covering the HKH region were mosaicked into a single DEM. For each lake polygon, elevation was approximated by sampling the DEM at the polygon centroid location. The sampled elevation value was assigned as the representative lake elevation.

Elevation Band Analysis: Based on the derived lake elevations, lakes were grouped into elevation bands of 500 m width: 3000–3500 m, 3500–4000 m, 4000–4500 m, 4500–5000 m, 5000–5500 m, 5500–6000 m, and > 6000 m. For each elevation band and inventory year, lake count and total lake area were calculated.

Inter-Annual Change Metrics: Changes between the 2020 and 2025 inventories were quantified using both absolute and relative metrics. The change in lake number (ΔN) and total lake area (ΔA) were computed as:

$$\Delta N = N_{2025} - N_{2020}, \tag{1}$$

$$\Delta A = A_{2025} - A_{2020}. \tag{2}$$

The percentage area change (PAC) was defined as:

$$\text{PAC} = \frac{A_{2025} - A_{2020}}{A_{2020}} \times 100\ (\%), \tag{3}$$

and the annual percentage area change (APAC) was calculated by normalizing PAC over the five-year interval:

$$\text{APAC} = \frac{\text{PAC}}{\Delta t}, \tag{4}$$

where $\Delta t = 5$ years.

Lake Matching and Change Classification: To assess lake-level evolution, polygons from the 2020 and 2025 inventories were spatially matched using an overlap-based approach. Candidate matches were identified using a spatial index, and polygon intersections were evaluated using an intersection-over-union (IoU)–based metric. Lakes were considered matched if their IoU exceeded a threshold of 0.01. Prior to intersection operations, invalid geometries were corrected using a zero-width buffer operation.

For matched lakes, relative area change was computed as:

$$\Delta A_{\text{rel}} = \frac{A_{2025} - A_{2020}}{A_{2020}}. \tag{5}$$

Based on relative area change, lakes were classified as *expanded* ($\Delta A_{\text{rel}} \geq 0.25$), *shrunk* ($\Delta A_{\text{rel}} \leq -0.25$), or *stable*. Lakes present in the 2025 inventory without a corresponding 2020 match were classified as *new*. These classifications represent operational definitions based on spatial overlap criteria.

DEM Reprojection and Slope Derivation: A DEM mosaic covering the study region was reprojected to the analysis CRS (EPSG:6933) using bilinear resampling to preserve continuous elevation values. Slope was then derived from the projected DEM in degrees using a block-wise implementation to limit memory usage for large rasters. Local elevation gradients were approximated using finite differences in the x and y directions, and combined into slope magnitude:

$$\text{slope} = \tan^{-1}\left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\right), \tag{6}$$

$$\text{slope}_{\text{deg}} = \text{slope} \cdot \frac{180}{\pi}. \tag{7}$$

E. PDGL Identification

For identifying potentially dangerous glacial lakes, we adapted the multi-factor scheme proposed by [10].

G11: Mean Slope of the Parent Glacier: For each lake, a "parent glacier" was assigned using a spatial index over glacier polygons. We first tested for glaciers intersecting the lake polygon; if multiple intersections existed, the glacier with the largest intersection area was selected. If no intersection existed, we selected the nearest glacier (based on centroid-to-centroid distance) within a maximum search distance (5 km). The G11 factor was then computed as the mean slope (degrees) over the selected parent glacier polygon:

$$G11 = \overline{\text{slope}}_{\Omega_{\text{parent glacier}}}. \tag{8}$$

R1: Steep Non-Glacierized Terrain Around the Lake: R1 measures steep terrain surrounding a lake that is *not* glacier-covered, intended to approximate potential avalanche/rockfall contributing slopes. For each lake polygon, we created a buffer of radius $R = 2000$ m and removed any glacier-covered portion within this buffer (glacier union difference). Within the resulting non-glacierized buffer zone, we counted slope raster pixels exceeding a steepness threshold of 30° and converted to area:

$$R1 = N_{>30^\circ} \cdot (r_x r_y) \cdot 10^{-6}, \tag{9}$$

where $N_{>30^\circ}$ is the number of pixels with slope $> 30^\circ$, and r_x, r_y are raster resolutions in metres (cell area $r_x r_y$). The result is reported in km^2.

D3: Approximate Moraine-Dam Slope Near the Outlet: D3 was implemented as an approximation of the dam-zone steepness near the lake outlet. For each lake boundary, we sampled $N = 100$ evenly spaced points along the polygon perimeter and extracted DEM elevations at these points. The point with the minimum sampled elevation was treated as a proxy for the outlet. We then buffered this outlet proxy by 200 m to define a dam-zone region Ω_{dam}, and computed:

$$D3 = \overline{\text{slope}}_{\Omega_{\text{dam}}}. \tag{10}$$

This provides a consistent, automated surrogate for dam steepness when explicit moraine-dam polygons are unavailable.

H7: Upstream Watershed Area: H7 represents contributing catchment area upstream of the lake, estimated using flow-routing on the projected DEM. For computational tractability, the DEM was subset to an area-of-interest (AOI) defined by the union of lake geometries plus a 5 km margin. Using `pysheds`, we filled sinks, computed flow direction, and delineated the catchment draining to the outlet proxy. Catchment area was computed as:

$$H7 = N_{\text{catch}} \cdot (r_x r_y) \cdot 10^{-6} \quad [\text{km}^2], \tag{11}$$

where N_{catch} is the number of catchment cells.

L9: Lake Perimeter: L9 was taken directly as the lake perimeter in kilometres:

$$L9 = P_{\text{lake}}. \tag{12}$$

Normalization, Weighting, and Hazard Index: Each factor was min–max normalized across all lakes (excluding NaNs and infinities). The composite hazard index (HI) was computed as a weighted sum of normalized factors:

$$HI = 0.27\, G11^{*} + 0.25\, R1^{*} + 0.21\, D3^{*} + 0.17\, H7^{*} + 0.10\, L9^{*}, \tag{13}$$

where missing factor values were conservatively filled with zero contribution (i.e., treated as $X^{*} = 0$) to allow HI computation for all lakes.

Hazard Classification and PDGL Flagging: We classified the continuous hazard index into five ordinal classes using Jenks natural breaks (`mapclassify.NaturalBreaks`, $k = 5$): {`very_low`, `low`, `medium`, `high`, `very_high`}. Lakes in the `high` and `very_high` classes were flagged as Potentially Dangerous Glacial Lakes (PDGLs):

$$\text{PDGL} = \begin{cases} 1, & \text{if class} \in \{\texttt{high}, \texttt{very\_high}\}, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

F. Estimation of population impacted by PDGLs

We estimated the number of people potentially exposed downstream of potentially dangerous glacial lakes (PDGLs) by summing population counts from a gridded population raster within cumulative distance bands (e.g., 5, 10, and 20 km).

III. Results

A. Performance of CNN and Transformer Models (RGB Dataset)

Table I summarizes the best-performing configurations for major architectures.

TABLE I
BEST RESULTS ON RGB DATASET

Model	backbone/hyp param	split	Loss	IoU	F1
U-Net	effnet-b0	train	0.3113	0.6326	0.7384
U-Net	effnet-b0	val	0.3087	0.6379	0.7423
DeepLabv3+	timm-effnet-b0	train	0.3432	0.6034	0.7159
DeepLabv3+	timm-effnet-b0	val	0.3231	0.6250	0.7348
U-Net++	timm-effnet-b0	train	0.2863	0.6543	0.7583
U-Net++	timm-effnet-b0	val	0.2884	0.6572	0.7587
U-Net++ Attn	timm-effnet-b0	train	0.3020	0.6422	0.7465
U-Net++ Attn	timm-effnet-b0	val	0.2910	0.6558	0.7566
MANet	timm-effnet-b0	train	0.3162	0.6446	0.7480
MANet	timm-effnet-b0	val	0.3020	0.6555	0.7602
SegFormer	timm-effnet-b0	train	0.2719	0.6686	0.7713
SegFormer	timm-effnet-b0	val	0.2889	0.6540	0.7582
TransUNet	h=8, d=6, mlp=5	train	0.6176	0.5121	0.6217
TransUNet	h=8, d=6, mlp=5	val	0.5710	0.5736	0.6879
Swin U-Net	mit-b2	train	0.4385	0.5273	0.6325
Swin U-Net	mit-b2	val	0.3740	0.5824	0.6888

B. Multi-Band Experiments (8-Band SR and RGB+NIR+NDWI)

Top three performing models from RGB experiments were trained on multibands but incorporating additional spectral bands did not improve model performance. Table II summarizes these results.

TABLE II
PERFORMANCE WITH MULTI-BAND INPUTS

Model	Bands	Split	Loss	IoU	F1
UNet++	8-band	train	0.3279	0.6178	0.7284
UNet++	8-band	val	0.3140	0.6361	0.7364
MANet	8-band	train	0.3172	0.6324	0.7415
MANet	8-band	val	0.3227	0.6371	0.7347
SegFormer	8-band	train	0.3250	0.6158	0.7298
SegFormer	8-band	val	0.3059	0.6440	0.7443
UNet++	RGB+NIR+NDWI	train	0.2958	0.6437	0.7513
UNet++	RGB+NIR+NDWI	val	0.3491	0.5946	0.7013
MANet	RGB+NIR+NDWI	train	0.2979	0.6423	0.7487
MANet	RGB+NIR+NDWI	val	0.3855	0.5625	0.6690
SegFormer	RGB+NIR+NDWI	train	0.2867	0.6508	0.7605
SegFormer	RGB+NIR+NDWI	val	0.3379	0.6042	0.7052

C. Final 20-Epoch Training (Top 3 Models)

Finally, extended training was done for top three models on RGB data (due to its best segmentation performance) that improved generalization. Table III presents the best results.

TABLE III
FINAL 20-EPOCH RESULTS ON RGB

Model	Split	Loss	IoU	F1
UNet++	train	0.2813	0.6609	0.7606
UNet++	val	0.2825	0.6652	0.7663
MANet	train	0.2318	0.7127	0.8043
MANet	val	0.2685	0.6770	0.7770
SegFormer	train	0.2143	0.7232	0.8186
SegFormer	val	0.2826	0.6607	0.7634

MANet achieved the highest IoU so, it was used in the final inference pipeline that successfully generated GIS-ready lake polygons from PlanetScope imagery.

D. Overall Changes in Glacial Lake Number and Area (2020–2025)

Across the HKH region, the 2025 inventory contains more lake polygons (N = 11,771) than the 2020 inventory (N =

8,808), yet exhibits a lower total lake area (99.920 km^2 versus 136.399 km^2). This corresponds to a net area change of $\Delta A = -36.479\ \mathrm{km}^2$, equivalent to a percentage area change (PAC) of -26.74% over the 2020–2025 period and an annual percentage area change (APAC) of $-5.35\%\ \mathrm{yr}^{-1}$.

E. Size-Class Dependent Changes in Glacial Lakes

Distribution of lakes with respect to size-class indicates that the increase in lake count is dominated by the smallest lakes ($< 0.01\ \mathrm{km}^2$; $+3401$ lakes), whereas net area reductions are concentrated in larger size classes, particularly the 1–5 km^2 class, which experienced a total area loss of $14.526\ \mathrm{km}^2$.

TABLE IV
CHANGES IN GLACIAL LAKE NUMBER AND AREA BY SIZE CLASS BETWEEN 2020 AND 2025.

Size class	$\mathbf{N_{2020}}$	$\mathbf{A_{2020}}$ (km^2)	$\mathbf{N_{2025}}$	$\mathbf{A_{2025}}$ (km^2)	ΔN	ΔA (km^2)
< 0.01	6752	13.987	10153	15.298	3401	1.311
0.01–0.05	1513	35.418	1188	27.330	-325	-8.089
0.05–0.1	302	20.868	249	17.235	-53	-3.633
0.1–0.5	218	39.168	170	30.558	-48	-8.610
0.5–1	14	10.017	9	7.085	-5	-2.933
1–5	9	16.941	2	2.414	-7	-14.526
5–10	0	0.000	0	0.000	0	0.000
> 10	0	0.000	0	0.000	0	0.000

F. Elevation-Band Distribution and Area Changes

Distribution of lakes with respect to elevation-band shows that lake area remains consistently concentrated between 4000 and 5000 m, with the largest area decrease occurring in the 4000–4500 m band ($-17.748\ \mathrm{km}^2$).

TABLE V
CHANGES IN GLACIAL LAKE NUMBER AND AREA BY ELEVATION BAND BETWEEN 2020 AND 2025.

Elev (m)	$\mathbf{N_{2020}}$	$\mathbf{A_{2020}}$ (km^2)	$\mathbf{N_{2025}}$	$\mathbf{A_{2025}}$ (km^2)	ΔN	ΔA (km^2)
$3k$–$3.5k$	459	5.856	1350	3.388	891	-2.468
$3.5k$–$4k$	2142	29.422	2792	22.149	650	-7.273
$4k$–$4.5k$	4089	63.556	3705	45.808	-384	-17.748
$4.5k$–$5k$	1835	28.071	2218	24.610	383	-3.461
$5k$–$5.5k$	207	1.547	221	0.993	14	-0.554
$5.5k$–$6k$	4	0.022	32	0.019	28	-0.004
$> 6k$	0	0.000	0	0.000	0	0.000

G. Lake-Level Change Classification Based on Polygon Overlap

Overlap-based polygon matching between 2020 and 2025 indicate a high fraction of operationally “new” lake features in 2025 ($N = 8876$). 1699 lakes are found to remain stable while 618 lakes are determined to get shrunk and 578 lakes have expanded.

H. Hazard Classes and PDGL Counts

Jenks natural breaks classification partitioned the hazard index into five ordinal groups. The resulting class counts were:

- `very_low`: 2,133 lakes (18.12%),
- `low`: 3,422 lakes (29.07%),
- `medium`: 3,369 lakes (28.62%),
- `high`: 2,017 lakes (17.14%),
- `very_high`: 830 lakes (7.05%).

I. PDGL exposure computation

Summing across all PDGLs yields the following cumulative exposed population totals:

TABLE VI
CUMULATIVE POPULATION EXPOSURE TOTALS WITHIN DISTANCE BANDS ACROSS ALL PDGLS

Distance band	Total population exposed
Within 5 km	8.236602×10^6
Within 10 km	4.501475×10^7
Within 20 km	2.978903×10^8

REFERENCES

[1] A. Basit and M. A. Siddique, “Hkh-pk-2020: Glacial lake inventory of the hindu kush-himalaya (pakistan) derived from planetscope imagery (september 2020),” May 2025. [Online]. Available: https://doi.org/10.5281/zenodo.15476013

[2] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” 2015. [Online]. Available: https://arxiv.org/abs/1505.04597

[3] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, “Unet++: A nested u-net architecture for medical image segmentation,” 2018. [Online]. Available: https://arxiv.org/abs/1807.10165

[4] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, B. Glocker, and D. Rueckert, “Attention u-net: Learning where to look for the pancreas,” 2018. [Online]. Available: https://arxiv.org/abs/1804.03999

[5] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, “Encoder-decoder with atrous separable convolution for semantic image segmentation,” 2018. [Online]. Available: https://arxiv.org/abs/1802.02611

[6] T. Fan, G. Wang, Y. Li, and H. Wang, “Ma-net: A multi-scale attention network for liver and tumor segmentation,” *IEEE Access*, vol. 8, pp. 179 656–179 665, 2020.

[7] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, and Y. Zhou, “Transunet: Transformers make strong encoders for medical image segmentation,” 2021. [Online]. Available: https://arxiv.org/abs/2102.04306

[8] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, “Swin-unet: Unet-like pure transformer for medical image segmentation,” 2021. [Online]. Available: https://arxiv.org/abs/2105.05537

[9] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, “Segformer: Simple and efficient design for semantic segmentation with transformers,” *Advances in neural information processing systems*, vol. 34, pp. 12 077–12 090, 2021.

[10] T. Zhang, W. Wang, T. Gao, B. An, and T. Yao, “An integrative method for identifying potentially dangerous glacial lakes in the himalayas,” *Science of the total Environment*, vol. 806, p. 150442, 2022.